Status	Finished
Started	Thursday, 17 September 2026, 5:33 PM
Completed	Thursday, 17 September 2026, 5:52 PM
Duration	19 mins 22 secs
Marks	143.00/146.00
Grade	**9.79** out of 10.00 (**97.95**%)

Question 1

Correct

Mark 28.00 out of 28.00

7 + 7 : bit calculation can be demonstrated as

4-Bit Signed-Integer Architecture:

0111 + 0111 = 1110

7 + 7 = -2

Evidence:

Carry → No evidence. Then Carry flag = 0 .

Borrow → No evidence. Then Borrow flag = 0 .

Postive overflow → Yes. Then Positive Overflow flag = 1 .

Negative overflow → No evidence. Then Negative Overflow flag = 0 .

Zero → No evidence. Then Zero flag = 0 .

4-Bit Unsigned-Integer Architecture:

0111 + 0111 = 1110

7 + 7 = 14

Evidence:

Carry → No evidence. Then Carry flag = 0 .

Borrow → No evidence. Then Borrow flag = 0 .

Postive overflow → No evidence. Then Positive Overflow flag = 0 .

Negative overflow → No evidence. Then Negative Overflow flag = 0 .

Zero → No evidence. Then Zero flag = 0 .

Question 2

Correct

Mark 29.00 out of 29.00

-4 + 2 : bit calculation can be demonstrated as

4-Bit Signed-Integer Architecture:

1100 + 0010 = 1110 → (in Decimal) -4 + 2 = -2

Evidence:

Carry → No evidence. Then Carry flag = 0 .

Borrow → No evidence. Then Borrow flag = 0 .

Postive overflow → No evidence. Then Positive Overflow flag = 0 .

Negative overflow → No evidence. Then Negative Overflow flag = 0 .

Zero → No evidence. Then Zero flag = 0 .

4-Bit Unsigned-Integer Architecture: -4 + 2 is **invalid**; if machine forces the bits:

1100 + 0010 = 1110 → in Decimal, unsigned interpretation: 12 + 2 = 14 .

Evidence:

Carry → No evidence. Then Carry flag = 0 .

Borrow → No evidence. Then Borrow flag = 0 .

Postive overflow → No evidence. Then Positive Overflow flag = 0 .

Negative overflow → No evidence. Then Negative Overflow flag = 0 .

Zero → No evidence. Then Zero flag = 0 .

But actually, -4 is a negative number that doesn't exist in an unsigned bit-architecture register.

Question 3

Partially correct

Mark 35.00 out of 37.00

2 - 4 : 4-bit calculation can be demonstrated as

4-Bit Signed-Integer Architecture:

0010 - 0100

Subtraction as addition: 2 + (-4)

0010 + 1100 = 1110 → then

0010 - 0100 = 1110 → (in Decimal) 2 - 4 = -2

Evidence:

Carry → No evidence. Then Carry flag = 0 .

Borrow → Yes. Then Borrow flag = 1 .

Postive overflow → No evidence. Then Positive Overflow flag = 0 .

Negative overflow → No evidence. Then Negative Overflow flag = 0 .

Zero → No evidence. Then Zero flag = 0 .

4-Bit Unsigned-Integer Architecture: 2 - 4 is -2; in 4-bit unsigned:

0010 - 0100 = 1110 → in Decimal, unsigned interpretation: 2 - 4 = 14 .

Evidence:

Carry → No evidence. Then Carry flag = 0 .

Borrow → Yes. Then Borrow flag = 1 .

Postive overflow → No evidence. Then Positive Overflow flag = 0 .

Negative overflow → No evidence. Then Negative Overflow flag = 0 .

Zero → No evidence. Then Zero flag = 0 .

Question 4

Correct

Mark 12.00 out of 12.00

10 - 3 : bit calculation can be demonstrated as

4-Bit Signed-Integer Architecture: 10-3 is invalid, since 10 is a/an positive overflow in the 4-Bit Signed-Integer Architecture.

4-Bit Unsigned-Integer Architecture:

1010 - 0011 = 0111 → (in Decimal) 10 - 3 = 7

Evidence:

Carry → No evidence.

Borrow → No evidence.

Postive overflow → No evidence.

Negative overflow → No evidence.

Zero → No evidence.

Question 5

Partially correct

Mark 23.00 out of 24.00

1 - 1 : bit calculation can be demonstrated as

4-Bit Signed-Integer Architecture:

0001 + 1111 = 0000

Evidence:

Carry → Yes.

Borrow → No evidence.

Postive overflow → No evidence.

Negative overflow → No evidence.

Zero → Yes.

4-Bit Unsigned-Integer Architecture: If A-B procedure was applied.

0001 - 0001 = 0000

Evidence:

Carry → No evidence.

Borrow → No evidence.

Postive overflow → No evidence.

Negative overflow → No evidence.

Zero → Yes.

4-Bit Unsigned-Integer Architecture: If no Subtraction instruction and two's complement can be applied if not exceed the 4-bit architecture.

0001 + 1111 = 0000

Evidence:

Carry → Yes.

Borrow → No evidence.

Postive overflow → Yes.

Negative overflow → No evidence.

Zero → Yes.

Question 6

Correct

Mark 16.00 out of 16.00

3 - 5 : bit calculation can be demonstrated as

4-Bit Signed-Integer Architecture:

0011 + 1011 = 1110

Carry → No evidence.

Borrow → No evidence.

Postive overflow → No evidence.

Negative overflow → No evidence.

Zero → No evidence.

4-Bit Unsigned-Integer Architecture:

0011 - 0101 = 1110

Since `the result` is outside the representable range of **4-bit unsigned integers** (0 to 15), the operation cannot represent the correct answer. This situation is handled by **wrapping around** modulo 16.

Carry → No evidence.

Borrow → Yes.

Postive overflow → No evidence.

Negative overflow → No evidence.

Zero → No evidence.